DEPARTMENT OF HEALTH SCIENCES
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COMPARISON OF BIOGAS PRODUCTION USING DIFFERENT DIGESTER OPERATIONAL STRATEGIES

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Abstract

Considerable challenges have accompanied previous anaerobic digesters leading to the development of a new cartridge design anaerobic digestion. Three methods (regarding different numbers of cartridges and days of rotation) were used to treat lignocellulosic biomass while generating a stable production of biogas for energy without floating issues and minimal liquid waste. Average methane yield was 7.57, 7.11, and 6.90 L/day/kg-VS. Microaeration maintained a low H_2S concentration at 60.48, 110.31, and 52.79 ppm. The study demonstrated the feasibility of using this new design for biogas production from lignocellulosic biomass while providing a baseline for system optimization.

Introduction

Cartridge design anaerobic digestion is a waste management method as well as an energy production technology. It breaks down organic matter in the absence of oxygen, and in the meantime, produces biogas which is a renewable energy source. There is a growing interest in using anaerobic digestion to treat lignocellulosic biomass such as crop residual and yard wastes. However, existing designs, including solid-state anaerobic digestion and liquid phase anaerobic digestion, have run into many problems, such as floating of biomass, effluent generation, unstable biogas production, and high H_2S and NH_4 concentrations, which give rise to challenges in system management and biogas utilization and may even cause digester failure. This study aimed to develop a novel system to resolve these challenges using a cartridge-design system.

	# of cartridges	Days for rotation
Method 1	3	7
Method 2	4	7
Method 3	4	9-10

Feedstock: Corn stover



Digestion Effluent, from a local waste water district



Reactor and cartridges Working volume: 4.0 L
Operated at 37°C





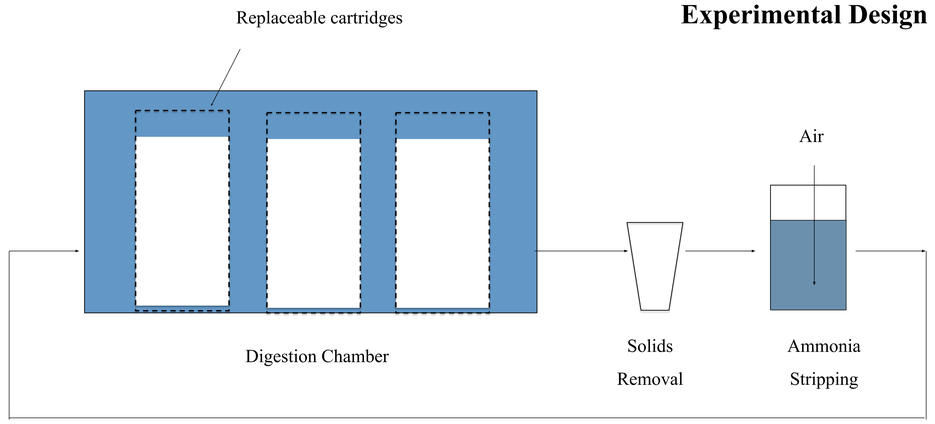
Composition of Gas

50-60% CH_4, 35% CO_2



Composition and volume are measured every 2-3 days.

Experimental Design



Biogas Applications

Biogas can be burned to provide heat. After cleaning and upgrading, biogas can be used for electricity generation or as a transportation fuel.

Methods

Analytical equipment:
Biogas: Landtec Biogas 5000
pH and alkalinity: Oakton 1100
NH_4-N: Spectrophotometer HACH DR2010
Volatile solids: Fisher Scientific ovens

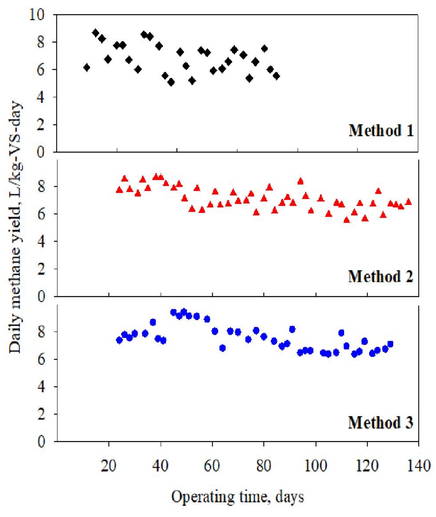
Statistical analysis: SigmaPlot and MS Excel

Results

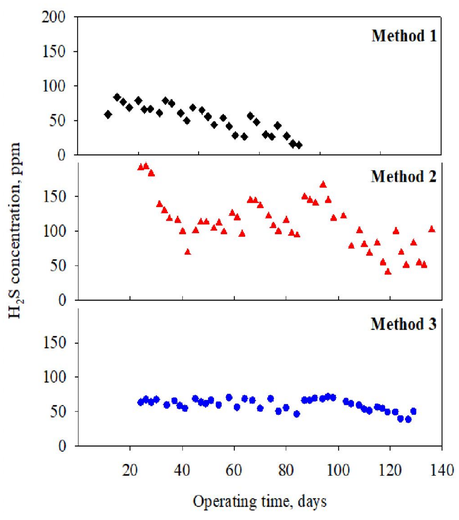
- Daily methane yield was stable for all three methods, with an average of 7.57, 7.11, and 6.90 L/day/kg-VS for methods 1, 2, and 3, respectively.
- The method 3 showed the highest cumulative methane yield of 259 L/kg-VS and the lowest H2S concentration of 52.79 ppm.

Results	Method 1	Method 2	Method 3
Avg CH4%	54.48	48.04	47.52
Avg CO2%	40.13	46.24	46.26
Avg O2%	1.98	1.22	1.34
Avg H2S ppm	60.48	110.31	52.79
Days/cycle	21	28	38
Accumulative methane yield per cycle, L/kg-VS	158.95	184.85	259.00

Daily methane yield comparison



H_2S conc. comparison



Conclusions

Compared the performance of three operation methods with varied number of cartridges and days of rotation.

All three methods showed stable biogas and methane production with no floating problem and minimal liquid wastes.

The method 3 showed the highest cumulative methane yield and the lowest H_2S concentration.

Future work

A manuscript will be submitted to a peer-reviewed journal and a proposal will be submitted to USDA.

References

1. Yang, L., D. Kopsell, A. Kottke, and M. Johnson. 2017. Development of a cartridge design anaerobic digestion system for lignocellulosic biomass. Biosystems Engineering. 160: 134-139.
2. L. Yang and S. Corsolini. 2019. A Case Study on Converting Organic Farm Waste Vegetables to Biogas Using a Cartridge Design Anaerobic Digester. Applied Biochemistry and Biotechnology. 189(2):638-646.

Acknowledgement

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